



Build an AI Chatbot with Amazon Bedrock



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ity of computing resources, such as servers, storage, and software applications, over the internet. e need for physical infrastructure and enabling faster and more efficient delivery of services.

ing up a super-organized digital storage system, kind of like a high-tech filing cabinet.

tainer in the cloud, provided by Amazon Web Services (AWS). Here's a simple way to understand it:

a gigantic filing cabinet. This cabinet can hold thousands of folders and files.

, and each drawer is an S3 bucket. You can create as many of these drawers (buckets) as you need.

e documents you put inside the drawer. They can be photos, videos, text files, or any other kind of



Introducing Today's Project

In this project, I'm going to create a Amazon Bedrock multi turn chatbot. I want to learn this because the new AWS AI developer roles require this!

Key tools and concepts

The key tools I used include Amazon Bedrock (Guardrails and Foundation Models). Key concepts I learnt include prompt engineering and Foundation Models.

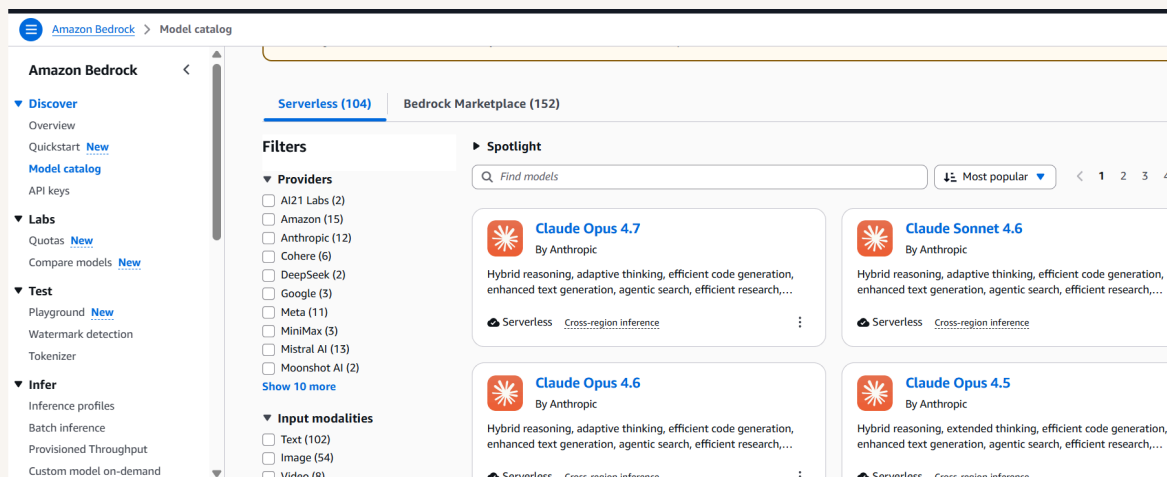
Challenges and wins

This project took me approximately 45 minutes including demo time!. The most challenging part was understanding Python script and temperature.



Discovering Amazon Bedrock

In this step, I'm going to navigate to Amazon Bedrock, explore foundation models and chat to the Amazon Nova 2 Lite model in the playground.



Understanding Foundation Models

I learned that a foundation model is pre-trained model on large datasets. Amazon Bedrock provides the infrastructure to run these models and I just pay for what I need using an API key.



Chatting with an AI Model

I sent a prompt about cloud computing, (i.e. Explain cloud computing in 3 sentences) The model responded by explaining what cloud computing was.

 Nova 2 Lite

Input: 103 | Output: 452 | Latency: 3941 ms

 Explain cloud computing in 3 sentences

 Reasoning
[REDACTED]

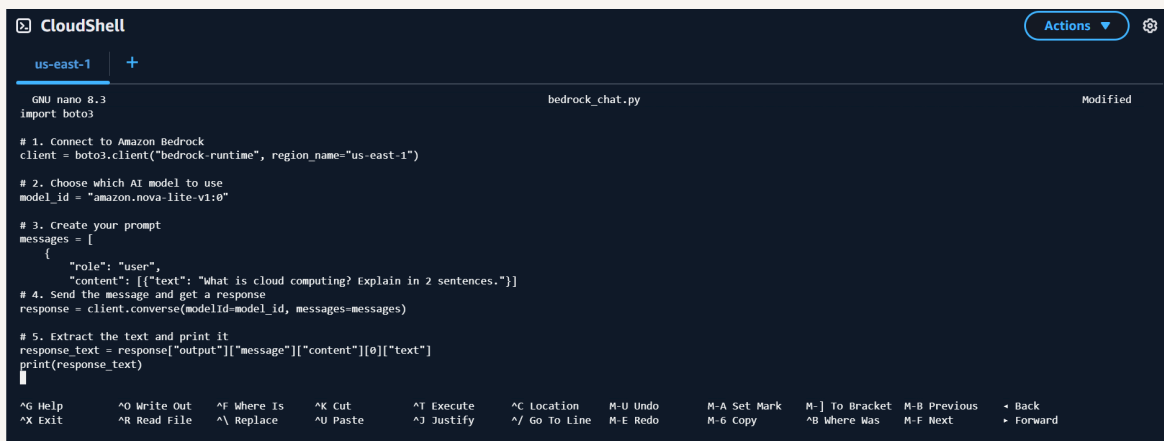
Cloud Computing Explained in 3 Sentences

- 1. Definition and Delivery:** Cloud computing refers to the delivery of various computing services—such as servers, storage, databases, networking, software, and analytics—over the internet ("the cloud") instead of relying on local servers or personal devices.
- 2. Access and Scalability:** Users can access these resources on-demand from anywhere with an internet connection, paying only for what they use, which allows businesses to easily scale capacity up or down based on their needs.
- 3. Benefits:** This model offers cost savings, reduced maintenance efforts, faster deployment of applications, and the flexibility to integrate advanced tools like AI and machine learning without managing underlying infrastructure.



Writing My First AI API Call

In this step, I'm going to use CloudShell to create a Python files and run these files to send a prompt to a foundation model.



```
CloudShell
us-east-1 +
GNU nano 8.3 bedrock_chat.py Modified
import boto3

# 1. Connect to Amazon Bedrock
client = boto3.client("bedrock-runtime", region_name="us-east-1")

# 2. Choose which AI model to use
model_id = "amazon.nova-lite-v1:0"

# 3. Create your prompt
messages = [
    {
        "role": "user",
        "content": [{"text": "What is cloud computing? Explain in 2 sentences."}]
    }
]

# 4. Send the message and get a response
response = client.converse(modelId=model_id, messages=messages)

# 5. Extract the text and print it
response_text = response["output"]["message"]["content"][0]["text"]
print(response_text)
```

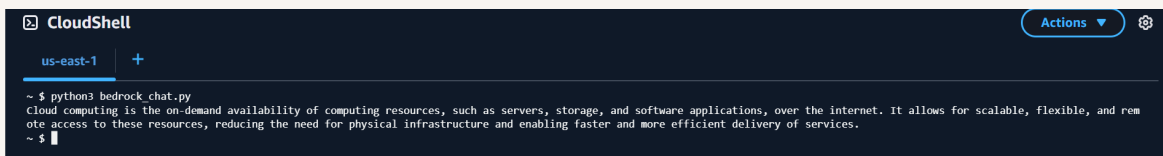
Understanding the Converse API

I created a client using boto3. The messages list contains role and content (hard coded prompt).



Running the Bedrock Script

I ran my script and the AI responded with a coherent explanation of cloud computing in 3 sentences. The response came from the Nova 3 Lite model.



```
CloudShell
us-east-1 +
~ $ python3 bedrock_chat.py
Cloud computing is the on-demand availability of computing resources, such as servers, storage, and software applications, over the internet. It allows for scalable, flexible, and remote access to these resources, reducing the need for physical infrastructure and enabling faster and more efficient delivery of services.
~ $
```



Building a Multi-Turn Chatbot

In this step, I am building a multi turn chatbot. Conversation history matters because it allows user to continue projects/ideas across different chats.

System Prompt and Inference Parameters

I configured the system prompt to be a friendly cloud computing tutor. I adjusted the inferenceConfig to include topP and temperature.

```
CloudShell
us-east-1 +
YOU: What is an S3 bucket?
Bot: Sure thing! let's imagine you have a huge, magical closet that can store anything you want-photos, videos, documents, or even your favorite books. This closet is so special that it can grow as big as you need it to be, and it's always accessible whenever you want to add or retrieve something.

In the world of cloud computing, an S3 bucket is like that magical closet, but in the cloud. S3 stands for "Simple Storage Service," and it's a service provided by Amazon Web Services (AWS). Here's how it works:

1. **Storage**: Just like your closet, an S3 bucket stores data. This data can be anything from text files, images, videos, to backups of your data.

2. **Organization**: Inside your magical closet, you might have different sections or shelves for different types of items. In an S3 bucket, you can organize your data into "folders" or "keys" (which are like file names) to keep things tidy.

3. **Accessibility**: Your magical closet can be accessed from anywhere, at any time, as long as you have the key (or password). Similarly, an S3 bucket can be accessed from anywhere in the world over the internet.

4. **Scalability**: If you need more space, your magical closet can expand without you having to do anything. An S3 bucket can also grow automatically to accommodate more data as you add more files.

5. **Durability**: Your magical closet is built to be very secure, ensuring your items are safe and sound. An S3 bucket is designed to be highly durable, with multiple layers of protection to keep your data safe.

So, in summary, an S3 bucket is a highly scalable, accessible, and secure storage solution in the cloud, much like a magical, ever-expanding, and always-accessible closet!
YOU: quit
Goodbye!
~ $
```



Adding Guardrails for Responsible AI

In this project extension, I'm adding Guardrails to make sure users cannot send harmful content to the foundation model.

my-chatbot-guardrail

Delete

Test

Guardrail overview

Edit

Name

my-chatbot-guardrail

ID

pk5z0trote0z

Description

-

Status

✔ Ready

KMS key

-

Create date (UTC+05:30)

May 24, 2026, 17:33

ARN

arn:aws:bedrock:us-east-1:789193817467:guardrail/pk5z0trote0z

Cross-Region inference

-

Testing Content Safety

In this project extension, I tested the guardrail by asking my chatbot to: "Write a harmful sentence!". This was rejected by my chatbot.



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```
System Error: type 'bot' is not a type
You: Write a harmful sentence!
Bot: Sorry, I can't respond in a way that might encourage or promote harmful content. It's important to promote positive and healthy interactions online and offline. If you have any questions or concerns about any topic, feel free to ask, and I'll do my best to provide helpful and informative answers. Remember, it's always important to treat others with kindness and respect.
You: █
```



Project Wrap-Up

I did this project today to learn how to set up Amazon Bedrock to run Foundation Models and design multi-turn chatbot inside Amazon Cloudshell. Another skill I want to learn is how to create my own AI Foundation Model.



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